RESEARCH RESUBMISSION / ENGINEERING ACTION PLAN

VLM-DexSafeDagger — RA-L Resubmission & Codex Revision Brief

Status: Second-version RA-L rejection; planning a substantially revised new submission
Prepared: 2 November 2026
Manuscript: `26-3841_02_MS.pdf` (RA-L submission 26-3841.2, received 25 September 2026)
Prior author response and supplement: `26-3841_02_AR.pdf` (29 pages)
Review evidence: latest Reviewer 29, Reviewer 31, and Associate Editor (AE) decision screenshots provided in the discussion
Audience: Author, supervisor, and Codex working with experiment scripts and raw evaluation logs

> **Purpose.** Convert the review history and our discussion into a reproducible, executable plan. This is a working document, **not** a ready-to-submit author response. Facts reported in the manuscripts are differentiated from calculations and hypotheses. No new experimental results are claimed.

1. Executive decision

Central recommendation: Keep *risk-aware teacher intervention for dexterous visuomotor distillation* as the main technical contribution. Treat superiority of the customized RL teacher-training recipe as **optional, not yet established**. Retain the claimed distinct visual-semantic contribution of the VLM only to the extent that controlled comparisons support it.

The AE acknowledged considerable improvement but rejected the second version because of four unresolved issues: (1) teacher-training procedure, (2) fairness of the DexTrAH baseline, (3) inconsistencies in reported results, and (4) evidence for VLM-guided adaptation. Reviewer 29 was broadly positive and asked for clarification and limited analysis. Reviewer 31 acknowledged improvements but continued to challenge the scientific support for the contribution claims. These are *interpretations of review tone*, not numerical acceptance probabilities. [S3]

Submission strategy: Do not merely polish text. First repair or explain all numerical inconsistencies; then clarify teacher-training reproducibility and baseline fairness; finally strengthen the VLM component's causal evidence and the statistical presentation of physical experiments.

Decisions to make before editing claims

1. **Teacher-RL contribution:** either (A) remove the independent superiority claim and present the recipe as a fully specified implementation choice, or (B) perform a fair retuned DexTrAH comparison at matched compute and evaluation conditions.
2. **VLM contribution:** if the VLM does not consistently beat strong LLM-only / adaptive non-VLM baselines over independent runs, narrow the claim to feasibility or empirically observed adaptation rather than essential semantic superiority.
3. **Primary outcome:** explicitly distinguish **unsafe exploration during simulation distillation** from **unsafe outcomes of frozen deployed RGB policies**. The teacher, Bellman critic, intervention gate, and VLM do *not* run on the physical robot in the reported evaluation. [S1, S2]

2. Source-of-truth map

Source	Content and where to look	Reliability/limitation
[S1] `26-3841_02_MS.pdf`	8-page Version 2 manuscript. Contributions p.2; method/Alg.1 pp.3–4; reward-term Table I p.4; experimental setup p.5; Fig.3/Table III p.6; physical results pp.7–8.	Submitted statements, not guaranteed consistent with raw logs.
[S2] `26-3841_02_AR.pdf`	Prior point-by-point responses pp. ii–xiii; supplement pp.1–6. Supplement A.1 actor/critic observations and A.2 termination conditions; C.1 critic evaluation; C.2 five-seed / 40-object evaluations; C.5 curriculum baseline.	Many reproducibility details were already supplied, but not exact reward formulas/weights.
[S3] Latest review screenshots	Reviewer 29: 10 comments, broadly favorable. Reviewer 31: M1–M4, m1–m3. AE: rejection report with four issues above.	Readable screenshot record; preserve exact wording when drafting the formal response.
[S4] Official RA-L author information	`https://www.ieee-ras.org/publications/ra-l/ra-l-information-for-authors/`	Policy page consulted 2 November 2026; verify again immediately before filing.

Important: Do not assume a formula, weight, seed, or dataset split exists merely because it appeared in chat. Verify against the actual experiment configuration, code commit, and archived logs. Source names and absolute repository paths have **not** been provided; Codex must discover them, not invent them.

3. Four major objections: exact action mapping

M1 — Teacher reward is not completely reproducible

Reviewer 31 asks: exact mathematical formulation and weights for all reward terms. Current Table I provides only name/purpose for `r_dist`, `r_curl`, `r_align`, `r_contact`, `r_near`, `r_goal`, `r_lift`, and `r_action`. [S1 pp.3–4]

Already supplied: actor/critic observation dimensions and individual fields in Supplement A.1; numerical unsafe-termination definitions (including 60-Hz checking and collision/contact conditions) in Supplement A.2. [S2 pp.1, xii]

Missing or requiring verification: exact term expressions, coefficients, clamping, reward timing, terminal/timeout bootstrap behavior, PPO setup, relevant control/action scaling, training seeds, domain randomization, and a mapping to the actual implementation. **Do not call the entire teacher RL procedure absent; say the reward formulation is incomplete.**

Action: Generate a code-derived reward-specification table and a compact mathematical subsection or appendix. Cross-reference every term to code and configuration. No new training is inherently required to resolve M1.

M2 — Comparison with DexTrAH-RGB is potentially unfair

Reviewer 31 asks: re-tune the DexTrAH/FABRICS-related baseline for the Tiangong-2 Pro + InspireHand hardware and compare under matched budgets, or moderate the claim that the authors' RL recipe is superior. The reviewer also asks for a clear recipe-to-recipe difference and notes that the claimed *safety-aware RL reward* lacks explicit unsafe-event penalties. [S1 pp.2, 6; S3]

Reason: a baseline controller tuned for another robot may underperform because of hardware transfer rather than RL-method quality. Comparison (a) vs (b), and (d) vs (e), therefore cannot independently prove algorithmic superiority if tuning is unequal.

Preferred route: remove “superior RL recipe” and “novel safety-aware RL reward design” as independent contributions unless substantively established. Describe the teacher as experimental infrastructure; specify it fully; hold its weights and teacher pool fixed across intervention comparisons (b)/(c)/(e)/(f)/(g)/(h).

Optional stronger route: hardware-retune baseline, disclose tuning search/equal compute and train/eval conditions, then directly isolate recipe effects and any explicit reward/termination safety contribution. This is more costly and should be justified separately.

M3 — Fig.3/Table III and reported mean statistics

Reviewer 31 asks: reconcile Fig.3 unsafe-rate curve ranking with Table III, reconcile mean definitions between Tables II and III, and clarify the object subset. These are high-priority credibility problems. [S1 p.6; S3]

Verified arithmetic discrepancies if 'Mean' is the simple four-object average:

Ablation	Unsafe rates shown for WindCart, Boot, ToolBox, Cow (%)	Shown mean (%)	Four-object arithmetic mean (%)
(b) Vanilla DAgger	13.5, 20.5, 19.0, 17.5	17.4	17.625
(c) Disagreement SafeDAgger	12.0, 19.5, 18.5, 16.0	16.7	16.500
(e) Fixed critic gate	7.5, 15.0, 15.5, 12.0	11.9	12.500
(f) VLM gate	6.0, 10.5, 11.0, 8.5	9.3	9.000
(g) LLM-only gate	7.5, 12.5, 13.0, 10.5	10.2	10.875
(h) Adaptive curriculum	7.0, 14.0, 13.5, 11.0	11.7	11.375

Promising hypothesis, NOT established: the displayed `Mean` column may instead be an **all-40-object aggregate**. In Supplement C.2, all-40-object results over five seeds exactly reproduce Table III's reported means for (b), (c), (e), (f): 17.4%, 16.7%, 11.9%, 9.3%. The caption for Table III, however, refers to four representative objects. Confirm the dataset, denominator and aggregation code for each number. Do not silently relabel until logs prove it. [S1 p.6; S2 pp.3–4]

Additional claim audit:

- "7.5 percentage points lower unsafe rate than disagreement-only SafeDAgger" does **not** equal 16.7% − 11.9% = **4.8 percentage points** in the reported aggregate table. It might refer to a different training-window metric; trace the provenance or revise the wording.
- "Further 4 percentage points" for the VLM also does **not** equal 11.9% − 9.3% = **2.6 percentage points** in that table. Trace its exact source (training average vs final evaluation vs subset).
- "Up to 80% early unsafe reduction" needs a specified time window, baseline denominator and exact aggregation method.
- Table IV(a) reports 14.7% seen-object unsafe rate; Table IV(b) shows 14.8% for the corresponding VLM method. Investigate whether this is rounding, dataset versioning, or a transcription error.
- Fig.3 shows **online training trajectories** while Table III may report **frozen-policy evaluation**. The two should not be expected to have identical numeric rankings without clearly matched definitions.

Caution on Reviewer 31's plot reading: the Fig.3 caption maps blue=Vanilla DAgger (b), orange=SafeDAgger (c), green=DexSafeDagger (e), red=VLM (f). The reviewer's stated ranking may be a legend/mapping misunderstanding; check the underlying data and caption before claiming the reviewer is wrong. [S1 p.6]

Action: Build a versioned metrics audit and regenerate every result table and figure from one documented log reader. Record *method*, *seed*, *object*, *rollout mode (training/frozen evaluation)*, *sample count*, *metric denominator*, *aggregation level and checkpoint/episode window*. Any mismatch must produce an explicit failed validation report, not be fixed by hand.

M4 — The distinctive VLM contribution is not sufficiently demonstrated

Reviewer 31 asks: demonstrate that VLM-guided threshold adaptation has incremental value, especially across different visual contexts/unsafe scenes, rather than only showing (e) vs (f). Reviewer 29 similarly asks whether the VLM helps with unseen *scene semantics*, not just unseen geometries. [S1 pp.6–7; S3]

Already implemented:

- (e) = fixed-threshold disagreement + Bellman critic.
- (h) = adaptive non-VLM performance curriculum; implementation in Supplement C.5.

- (g) = language-model threshold tuner without image inputs.
- (f) = full VLM threshold tuner with images and rollout statistics. [S1 p.6; S2 pp.5–6]

Reported (not yet re-audited) Table III mean unsafe rates: (e) 11.9%; (h) 11.7%; (g) 10.2%; (f) 9.3%. Hence (f) vs (e) = 2.6 pp, vs (h) = 2.4 pp, vs (g) = 0.9 pp, according to the displayed means. The small *image-specific* (f) vs (g) difference is the key evidence gap.

Existing gap: Supplement C.2's five-seed analyses cover (b), (c), (e), (f), but **not (g) or (h)**. [S2 pp.3–4]

Required high-value study: repeat (g) and (h), matching (f)'s seeds and training/evaluation protocol; report mean, standard deviation, paired differences and confidence intervals by seed. Preserve teacher weights, student initialization logic, training samples, threshold-update cadence and total rollout budget as closely as possible.

Useful targeted controls: (i) correct images vs shuffled/irrelevant images with identical rollout statistics and prompt structure; (ii) compare results at *matched intervention budget* (mean beta), or show safety–autonomy tradeoff curves; (iii) analyze predefined visual-context groups (e.g., occlusion/clutter/appearance) with sufficient repeated conditions. These are **recommendations**, not experiments that the reviewers explicitly mandated by name.

Interpretation boundary: The VLM is a simulation-distillation-time threshold advisor queried roughly ten times per run, not a physical runtime safety monitor. Real-world unseen geometry performance alone does not establish VLM semantic generalization. [S1 p.5; S2 pp. iii–iv]

4. Reviewer 29: ten requested follow-ups

ID	Request	Already present?	Next action
R29.1	Maintain empirical, not formal, safety language	Mostly	Audit title, abstract, discussion, conclusion for guarantees.
R29.2	State ~39% unsafe rate on unseen physical objects	Data yes	Prominent and explicit limitation.
R29.3	Statistical significance of physical (e) vs (f)	Per-group rates reported	Paired or hierarchical confidence intervals / tests using actual trial pairing.
R29.4	Consequences of 5.8% critic false negatives	Rate supplied	Threshold sweep: false negatives, takeover frequency, lift success.
R29.5	Critic evaluation level and lead time	Episode-level denominator defined	Compute warning lead-time distribution and before-event detection.
R29.6	Teacher itself can be unsafe	Teacher average unsafe ~11.8% shown	Conditional teacher-takeover failures; clarify OR gate handles two *student-risk signals*, not teacher safety.
R29.7	Complete VLM reproducibility	Prompt/model/update details in S2	Check exact version, decoding, image selection, latency, failure handling, cost against runs.
R29.8	Semantic vs geometric generalization	Unseen geometric objects shown	Narrow claim or add scene-semantic stratification.
R29.9	Sim-only privileged vs hardware-observable unsafe events	Sim definitions in S2	Sensor/estimator measurement table for each unsafe predicate during physical eval.
R29.10	Compact limitations paragraph	Partial	Sim-to-real contact mismatch; external VLM; task-specific risk definitions; no formal guarantee.

Key existing numbers (must be checked against raw logs): critic false-negative rate = 5.8% of unsafe episodes in a passive, frozen-student evaluation; real-world seen unsafe rate = 14.7% and unseen = 39.2% in pooled evaluation; real-world fixed (e) vs VLM (f) unsafe rates = 16.1% vs 14.8% for seen and 41.4% vs 39.2% for unseen. [S1 pp.7–8; S2 pp.iv–vi]

5. Experimental and statistical protocol

Primary analysis units

- **Independent training seed** is the primary unit for variability across independently trained student policies. Avoid treating thousands of correlated episodes as thousands of independent *training* replications.
- **Object** is a clustering/generalization unit; distinguish seen versus unseen objects and report both macro-object means and episode-pooled rates when useful.
- **Episode** is the unit for binary lift success and any-unsafe-event rate. A single episode is unsafe if any defined event occurs; failure-mode decomposition has a specified priority when events coincide.
- **Time step** is the unit for intervention gate actions; beta = proportion of eligible control steps with teacher takeover. Be explicit about denominator and any takeover hold intervals.
- **Training window vs final frozen-policy evaluation** must never be mixed in one metric without a label.

Proposed metric output schema

One row per evaluation episode, at minimum:

```
method_id, training_seed, evaluation_seed, object_id, seen_unseen,
experiment_phase, checkpoint_step, evaluation_episode_id,
lift_success, unsafe_any, unsafe_mode, episode_steps,
teacher_steps, total_control_steps, first_warning_step,
first_unsafe_step, teacher_active_at_unsafe
```

Some fields may be unavailable and must be reported as missing rather than invented. Retain raw log paths, source hash, and calculation script commit for traceability.

Pre-specified statistical comparisons

1. Simulation: (f)−(g), (f)−(h), (f)−(e), and (e)−(c); paired by training seed and object when legitimate. Report effect direction with CI, not only isolated p-values.
2. Real: (f)−(e) on the same objects and protocols; use trial-level pairing only if individual conditions are genuinely matched. Otherwise use object/seed-clustered uncertainty estimates.
3. Warning system: critic threshold sweep for miss rate / warning lead time / beta / lift success. Note that the 5.8% value is *per unsafe episode*, not automatically a stepwise error rate.
4. Guard against cherry-picking: define primary metrics, conditions and acceptable seed sets before comparing plots. Correct/qualify claims even when results are weaker than expected.

6. Codex implementation plan (ordered work packages)

P0-A — Repository discovery and reproducibility inventory

Inputs: paper PDFs, source code repository, configurations, saved models, seeds, experiment logs and generated figures (if available).
Do: locate reward functions, termination code, teacher/critic/student training configs, per-method evaluation entry points, and plotting scripts; record code revision, run identifiers, missing artifacts and reproducible commands.
Deliver: `audit/inventory.md` plus a machine-readable run manifest.
Done when: every published table/plot row can be mapped to one or more source runs, or is explicitly marked as untraceable.

P0-B — Numeric results audit

Do: implement one deterministic evaluation aggregator; validate four-object means vs 40-object macro means; recompute each seed, object, and pooled rate; regenerate Table II/III/IV and Fig.3/4 where data exists; emit discrepancy report including “7.5 pp”, “4 pp” and “80%” claims.
Deliver: reusable analysis module, verification tests, `audit/metric_reconciliation.csv`, regenerated tables/figures.
Done when: every displayed mean is numerically consistent with its caption, population and calculation procedure; figure data origin is documented; no manual corrections remain.

P0-C — Teacher reward and baseline note

Do: extract reward math, coefficients and control limits **from code**; compare recipe differences with DexTrAH-inspired baseline; identify whether any explicit reward term depends on unsafe predicate; supply a precise experimental description.
Deliver: `audit/teacher_reward_spec.md`, recipe difference table, proposed revised text.
Done when: reward/termination descriptions independently reproduce the implemented logic; team selects retain-vs-remove RL superiority claim.

P1-A — Stronger VLM ablations

Do: if logs are missing, design and run matched experiments for (g) and (h) using the same independent seeds as (f); optional shuffled-image control; validate matching teacher weights, total training samples, update cadence and comparable intervention budget.
Deliver: `experiments/vlm_ablation_protocol.md`, manifests and raw results; reproducible comparison tables and uncertainty.
Done when: (f) vs (g)/(h)/(e) can be compared with transparent seed-level uncertainty and visual-input causality is tested or claims are narrowed accordingly.

P1-B — Physical statistical reanalysis

Do: inspect whether physical trials can be genuinely paired; compute difference in unsafe-event rate and lift success by seen/unseen and object; produce appropriate CIs / statistical tests, accounting for clustering.
Deliver: `audit/physical_statistics.md`, per-object effect plots, tables and code.
Done when: physical outcome claims explicitly match statistical support; intervention frequency is discussed only for simulation training.

P2-A — Critic and teacher safety diagnostics

Do: determine whether warnings precede failures; compute lead times; sweep tau_Q; quantify false-negative/teacher takeover conditional outcomes; distinguish detection from prevention.
Deliver: `audit/critic_warning_diagnostics.md` with denominators, curves and edge-case definitions.
Done when: Reviewer 29 comments 4–6 can be answered using measurable quantities, or limitations are disclosed.

P2-B — Manuscript, supplement and response editing

Do: update contributions and cautious safety wording; include exact reward math in manuscript/supplement as permitted; clarify what uses simulation privileged signals vs real measurements; finalize VLM prompt/decoding/model settings and resource costs; prepare a response mapping each AE and reviewer comment to manuscript section and evidence.
Deliver: clean paper PDF, marked-up version, response document, cross-reference index and validation checklist.
Done when: no unsupported claim survives; all claims have a source/result pointer; new submission requirements are met.

7. Suggested issue/task checklist for Codex

- [ ] Trace raw sources for every Table III entry (all eight methods) and document all aggregation dimensions.
- [ ]

Verify whether Table III `Mean` = 40-object pooled/macro mean or mean of four displayed objects.

☐ Recalculate all advertised absolute and relative unsafe-rate reductions from named comparable metrics.

☐ Export exact reward equations and weights from configuration and code, with references.

☐ Explicitly enumerate differences between the two teacher training recipes, including controller tuning.

☐ Confirm five-seed coverage of (b), (c), (e), (f), and assess whether new (g)/(h) runs are needed.

☐ Generate seed-level, object-level uncertainty for the VLM vs LLM-only and curriculum comparisons.

☐ Consider a shuffled-image VLM control and matched-beta safety–autonomy curves.

☐ Analyze real-world (e)/(f) effects using defensible paired or hierarchical statistics.

☐ Compute critic warning lead time and threshold tradeoffs; document unsafe teacher intervention cases.

☐ Draft concise limitations and remove suggestions of formal or deployment-time VLM safety guarantees.

☐ Assemble a new-submission response plus highlighted manuscript and cite the previous paper ID.

8. Codex operating constraints

Do not fabricate data. If plots, reward settings or run logs are missing, stop that item and report missing inputs. **Do not silently change training parameters** to force numerical agreement. **Do not launch expensive GPU experiments without user authorization**; first provide commands, resource budgets, planned seeds, conditions and checkpoints. **Do not delete original logs or overwrite old figures**; write generated outputs to an audited location. **Do not confuse teacher training, intervention-time simulation training, frozen simulation evaluation and deployed physical evaluation.**

Every research claim must carry its measurement population and time window. Always show the denominator for a percentage, especially 5.8% false negatives and reported unsafe-rate changes. Use the raw source for equations; keep implementation and paper synchronized.

9. RA-L new-submission packaging (policy check)

A rejected paper is handled as a **new submission** rather than the timed revise-and-resubmit path. RA-L's published policy requires disclosing the previously rejected submission number in Paperplaza and uploading **one PDF containing (1) the author response addressing editor/reviewer comments and (2) the manuscript with changes highlighted**. The previous paper title and number should be stated in the response. The normal 30-day deadline applies to an RR decision, not to submitting a new version of a rejected paper. Re-check the official policy and upload limits before filing. [S4]

For this paper, identify prior submission **26-3841** (including Version 2's rejection), make the new response self-contained, and focus on the latest AE/Reviewer 29/Reviewer 31 comments while documenting how earlier-round issues were resolved. There is no guarantee of the same editor/reviewers.

10. Final acceptance gates for the team

Gate	Required evidence	Decision if not met
G0: Reproducible numbers	Tables/plots regenerate and population/mean definitions are unambiguous.	No resubmission yet.
G1: Defensible teacher setup	Exact reward/config/termination information; baseline claim either fairly tested or withdrawn.	Remove superiority/safety-aware reward novelty claim.

Gate	Required evidence	Decision if not met
G2: Incremental VLM evidence	Matched comparisons, uncertainty, and at least one convincing visual-information test or narrower claim.	Reframe paper around critic-guided intervention.
G3: Physical honesty	Real comparisons with uncertainty; ~39% unseen-object unsafe rate prominently discussed.	Avoid claiming deployment-grade safety.
G4: Complete review response	Every latest major/minor comment mapped to evidence and changed text; official upload format.	Do not file until complete.

Reference documents

- **[S1]** `26-3841_02_MS.pdf`, Version 2 manuscript, eight pages, provided by the author.
- **[S2]** `26-3841_02_AR.pdf`, previous author response + supplement, 29 pages, provided by the author.
- **[S3]** Screenshots of Version 2 Reviewer 29 and Reviewer 31 reports and the AE rejection summary, provided in the chat.
- **[S4]** IEEE RA-L, *Information for Authors*, section “Submitting a previously rejected paper”: https://www.ieee-ras.org/publications/ra-l/ra-l-information-for-authors/ (consulted 2 November 2026).

Prepared from the submitted evidence and the discussion as a reference for revision work. Recommendations are proposed actions, not new findings or editor instructions.